

Experiment 1: DDL Commands – Exercise

Exercise 1: Create a Database

Write an SQL command to create a database with your ROLLNo

```
MariaDB [(none)]> CREATE DATABASE 24c029;  
Query OK, 1 row affected (0.002 sec)
```

Create a table named Student with the following attributes:

Column Name	Data Type	Constraint
Student_ID	INT	Primary Key
Name	VARCHAR(50)	NOT NULL
Gender	CHAR(1)	
Age	INT	
Department	VARCHAR(30)	
Email	VARCHAR(50)	

```
MariaDB [24co29]> create table Student (Student_ID int primary key, Name varchar(50)  
NOT NULL, Gender char(1), Age int, Department varchar(30), Email varchar(50));  
Query OK, 0 rows affected (0.037 sec)
```

```
MariaDB [(none)]> use 24co29;  
Database changed
```

Exercise 3: Display Table Structure

Display the structure of the Student table using the appropriate DDL command.

```
MariaDB [24co29]> describe student;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int(11)	NO	PRI	NULL	
Name	varchar(50)	NO		NULL	
Gender	char(1)	YES		NULL	
Age	int(11)	YES		NULL	
Department	varchar(30)	YES		NULL	
Email	varchar(50)	YES		NULL	
Mobile_No	varchar(15)	YES		NULL	

```
7 rows in set (0.034 sec)
```

Exercise 4: Add a New Column

Add a new column Mobile_No of type VARCHAR(15) to the Student table.

```
MariaDB [24co29]> Alter table student add(Mobile_No VARCHAR(15));
Query OK, 0 rows affected (0.023 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 5: Add Multiple Columns

Add the following columns to the Student table:

Address – VARCHAR(100)

Date_of_Birth – DATE

```
MariaDB [24co29]> Alter table student add(Address VARCHAR(100), Date_of_Birth DATE);
Query OK, 0 rows affected (0.026 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 6: Add a Column with Default Value

Add a column City of type VARCHAR(30) with the default value ''Goa'.

```
MariaDB [24co29]> Alter table student add(City VARCHAR(30) default 'Goa');
Query OK, 0 rows affected (0.006 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 7: Modify a Column

Modify the size of the Address column from VARCHAR(100) to VARCHAR(200).

```
MariaDB [24co29]> Alter table student modify Address VARCHAR(200);
Query OK, 0 rows affected (0.028 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 8: Rename a Column

Rename the column Mobile_No to Phone_Number.

```
MariaDB [24co29]> Alter table student change Mobile_No Phone_Number varchar(15);
Query OK, 0 rows affected (0.026 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 9: Drop a Column

Delete the City column from the Student table.

```
MariaDB [24co29]> Alter table student drop column City;
Query OK, 0 rows affected (0.009 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Exercise 10: Rename a Table

Rename the Student table to Student_Details.

```
MariaDB [24co29]> rename table Student to Student_Details;
Query OK, 0 rows affected (0.010 sec)
```

Exercise 11: Truncate a Table

Remove all records from the Student_Details table without deleting its structure.

```
MariaDB [24co29]> truncate table Student_Details;  
Query OK, 0 rows affected (0.037 sec)
```

```
MariaDB [24co29]> select * from Student_Details;  
Empty set (0.001 sec)
```

Exercise 12: Drop a Table

Delete the Student_Details table permanently.

```
MariaDB [24co29]> drop table Student_Details;  
Query OK, 0 rows affected (0.028 sec)
```

```
MariaDB [24co29]> select * from Student_Details;  
ERROR 1146 (42S02): Table '24co29.student_details' doesn't exist  
MariaDB [24co29]> |
```